

國立暨南國際大學資訊工程學系學士班必選修科目一覽表

Undergraduate Program Course Catalog of Department of Computer Science and Information Engineering, National Chi Nan University

本表(112)學年度入學新生適用 Effective for students admitted in Fall, 2023

最低畢業學分：128 學分，全校共同課程 15 學分，通識領域課程 16 學分

系必修 49 學分，專業選修 42 學分，自由學分 6 學分。 Undergraduate requirements: a minimum of 130 required credits, including 15 credits in the School-Wide Curriculum, 16 credits of General Education Courses, 49 credits of Departmental Required Courses, 42 credits of Elective Courses, and 6 credits of School Elective Courses.

壹、全校共同課程 1.School-Wide Curriculum			
一、共同課程（必修15學分） A. School-Wide Required Courses (15 credits)			
課程代號 Course Code	課程名稱 Course Name	學分數 Credits	備註 Notes
	英文一(上) English I(1)	2	自110學年度起，英文能力分班改為依照本校自辦英檢(OOPT)作依據，分為三級上課：前10%的同學將分配至「進階班英文」上課；中間80%的同學將分配至原班英文上課；後10%的同學將分配至「基礎班英文」上課。 Since the first semester of 110 Academic Year (September2021), the competence-based class grouping policy of Freshmen English (Course Title: English 1) will be based on registered student's OOPT scores; Students will be accordingly divided into three different levels: Advanced English (top 10%), General English (middle 80%) and Basic English (bottom 10%).
	英文一(下) English I(2)	2	
	英文二 English II	2	
	中文思辨與表達一：閱讀書寫 Critical Thinking and Expression In Chinese (I) : Reading And Writing	2	僑外生得以修習「僑外生華語文」抵免「中文思辨與表達」課程。(Overseas students are allowed to take the course 'Chinese for Overseas Students' as an exemption for the course 'Critical Thinking and Expression in Chinese'.)
	中文思辨與表達二：多元應用 Critical Thinking and Expression In Chinese (II) : Multiple Applications	2	
	社會服務學習(上) Service Learning(1)	1	依所屬學系分班上課，無特殊原因不得跨班選課。(Students are divided into groups by their own department, without any proper reasons they cannot change the decision.)
	社會服務學習(下) Service Learning(2)	1	
	大一體育(上)Physical Education (1)	0.5	上學期依所屬學系分班上課，無特殊原因不得跨班選課，惟未達40人之學系、院學士班班級，則採合併班開課；下學期則依運動項目分班，自由選課。(In the first semester of freshman, students are arranged to take Physical Education (1) by their original department. Without any proper reasons, change will be prohibited. If a class of the department or the interdisciplinary program that is less than 40 students, the combined class will be adopted. In the second semester of freshman, students are able to take Physical Education (2) by their interesting.)
	大一體育(下)Physical Education (2)	0.5	

特色運動 (Featured Sports)	2	大二開始，每學期限修一項，畢業前須完成至少2項特色運動 (Students are limited take one course each semester since Sophomore, At least 2 featured sports courses are required before graduation.)
通識講座 General Education lecture program	0	畢業前至少應出席6場。(Students are required to attend at least 6 lectures before graduation.)

二、共同選修課程 (各系依共同選修加列課程) B. School-Wide Elective Courses

第二外語 (Second Foreign Language)		含日文、韓文、法文、德文及西班牙文等。(Including Japanese, Korean, French, German and Spanish, etc.)
東南亞語言(Southeast Asia Languages)		含泰語、越南語、緬甸語、印尼語及馬來語，可抵特色通識「國際連結」次領域。(Including Thai, Vietnamese, Burmese, Indonesian and Malay. These courses can be credited course credits in International Connection of Characteristic General Education)
進階語文(Advanced English)		
其他選修(Other Elective Course)		

貳、通識領域課程 (至少16學分) 2.General Education Courses (16 credits)

主領域 Main Field	次領域 Sub-Field	學分數 Credits	備 註 Notes
人文領域 Humanities	文學與藝術(Literature and Art)	4	至少修習4學分。(take at least 4 credits)
	歷史哲學與文化(History, Philosophy and Culture)		
社會領域 Society	法政與教育(Law, Politics and Education)	4	至少修習4學分。(take at least 4 credits)
	社經與管理(Society, Economy and Management)		
自然領域 Nature	工程與科技(Engineering and Technology)	4	至少修習4學分。(take at least 4 credits)
	生命與科學(Life and Science)		
特色通識領域 Characteristic General Education	國際連結(International Connection) (東南亞Southeast Asia)	4	至少修習4學分。(take at least 4 credits)
	淨零永續(Net-Zero Sustainability)		
	(綠概念Green Concept)		
	社會創新(Social Innovation) (在地實踐Social Practice)		

通識各主領域課程(人文、社會、自然及特色通識)得單一領域跨域認抵，至多4學分 (需經通識教育中心認定之)
General education courses from various fields (humanities, society, nature, and characteristic general education) can be recognized across fields. A maximum of 4 credits is admitted in a single field. It must be approved by the General Education Center.

參、系必修 (至少49學分) 3.Departmental Required Courses (49 credits)

課程代號 Course Code	課程名稱(中文及英文) Course Name (Chinese and English)	學分數 Credits	修別 Course type (required/elective)	開課年級 Course availability	備 註 Notes
210036	計算機概論 Introduction to Computer Science	3	必	一上	
210003	微積分 (上) Calculus(1)	3	必	一上	

210013	離散數學 Discrete Mathematics	3	必	一上	
210008	微積分（下） Calculus(2)	3	必	一下	
210099	程式設計 Programming	3	必	一下	增修PHP網頁程式設計
210151	邏輯設計與實驗（一） Logic Design and Laboratory Experiments (1)	3	必	一下	
210111	機率 Probability	3	必	一下	
210016	資料結構與演算法（一） Data Structures and Algorithms (I)	3	必	二上	
210152	邏輯設計與實驗（二） Logic Design and Laboratory Experiments (2)	3	必	二上	
210022	資料結構與演算法（二） Data Structures and Algorithms (II)	3	必	二下	
210026	線性代數 Linear Algebra	3	必	二下	
210025	計算機組織與結構 Computer Organization and Architecture	3	必	三上	
210093	系統程式 System Programming	3	必	三上	
210121	專題（一） Project Study (I)	2	必	三上	
210122	專題（二） Project Study (II)	2	必	三下	
210102	作業系統 Operating Systems	3	必	三下	
210160	微電腦實驗 Microcomputer Systems Lab	3	必	三下	

肆、專業選修（至少42學分） 4.Elective Courses (42 credits)

210009	基本電學 Basic Electricity	3	選	1	
210012	資訊系統與網路導論(二) Introduction to Computer System and Network II	3	選	1	
210015	數位電子學 Digital Electronics	3	選	1	
210021	組合數學 Combinatorial Mathematics	3	選	1	
210023	數位電路實驗 Digital Circuits Lab	1	選	2	
210123	自動機與形式語言 Automata and Formal Language	3	選	3	
210124	數位系統設計 Digital System Design	3	選	1	
210125	數值方法 Numerical Methods	3	選	1	

210126	電腦視覺與人機介面 Computer Vision and Man-machine Interface	3	選	1	
210130	具體數學 Concrete Mathematics	3	選	1	
210131	UNIX程式設計 UNIX Programming	3	選	1	
210132	程式語言 Programming Languages	3	選	1	
210133	隨機程序 Introduction to Fundamental Stochastic Processes	3	選	2	
210134	LISP程式設計 LISP Programming	3	選	1	
210135	UNIX使用入門 Introduction to Unix Environment	3	選	1	
210136	電腦網路通信協定 Computer Network Protocols	3	選	1	
210140	智慧型行動裝置軟體設計 Intelligent Mobile Device Programming	3	選	1	
210141	3D建模與互動多媒體應用 3D Modeling and Interactive Multimedia Applications	3	選	1	
210142	數值分析 Numerical Analysis	3	選	1	
210143	Web服務和應用 Web Services and Applications	3	選	1	
210146	基礎程式設計 Computer Fundamentals &	1	選	1	
210147	基礎程式設計 (二) C++ Programming Fundamentals	1	選	1	
210148	專業實習 Specialized Internship	3	選	4	
210149	企業實習 Enterprise Internship	6	選	4	
210009	電子電路 Electrical Circuits	3	選	1	
210086	工程數學 Engineering Mathematics	3	選	2	
210115	邏輯設計 Logic Design	3	選	1	
210145	計算機網路 Computer Networks	3	選	1	
210029	微算機系統 Microcomputer Systems	3	選	1	
210153	MFC視窗程式設計 Windows MFC Programming	3	選	1	需先修程式設計
210154	個人電腦概論 Information Technology Essentials	3	選	1	需先修計算機概論
210156	JavaScript程式設計 JavaScript Programming	3	選	1	需先修程式設計

210157	企業實習240(一) Enterprise Internship 240(一)	3	選	4	
210158	企業實習240(二) Enterprise Internship 240(二)	3	選	4	
210159	企業實習(二) Corporate Internship (2)	6	選	4	
210161	Linux系統管理 Linux System Administratio	3	選	1	
210163	電腦與資訊安全概論 Introduction to Computer and Information Security	3	選	1	
210164	音樂,大腦,與科技 Music,Brain,Science and Technology	2	選	1	
210165	英文寫作, 口說, 簡報 (一) On English Writing, Speaking andPresentation (1)	3	選	1	
210166	數位音樂學習與創作 Digital Music Learning and Creation	2	選	1	
210167	Python 網路程式設計 Python Network Programming	3	選	2	
210168	程式競賽練習 Practice of Competitive Programming	3	選	1	
備註: 專業選修42學分, 3年級以上可選修系上碩士班課程作為專業選修。					